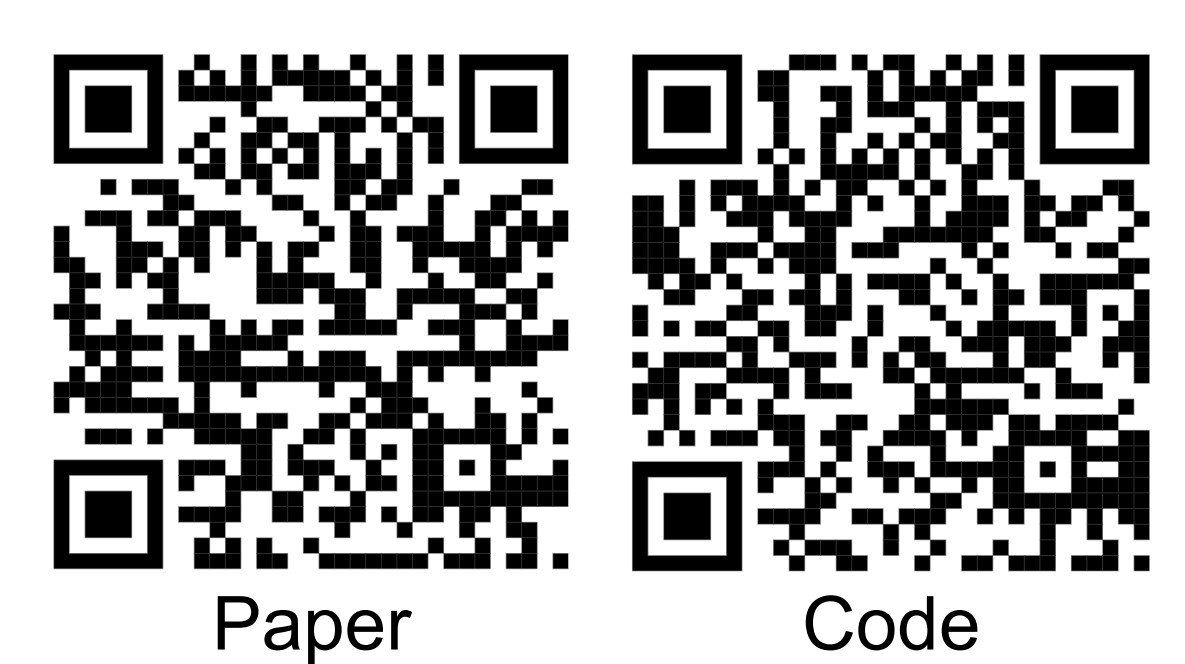


Layer-Centric Factors of Variation Disentanglement for Task- and Model-Agnostic Generalization

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Motivation

Disentanglement Learning

- Identifying and Separating the underlying Factors of Variations (FoV) for **model generalization**.

The Gap in Latent-Vector-Centric Methods

- Conventional disentanglement methods impose objectives only at the latent bottleneck.
- Improved intrinsic metrics do not reliably translate into downstream generalization gains.
- Downstream predictions are shaped by intermediate features throughout the network.

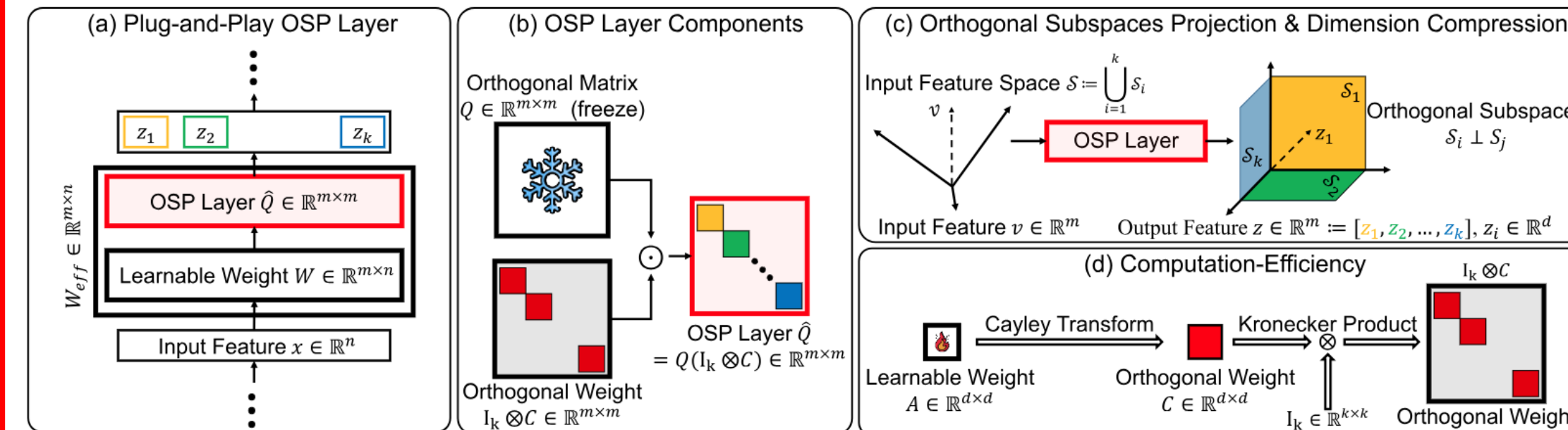
What We Target?

- What?: Disentanglement should be induced at the layer level, not just the latent space.
- How?: Given intermediate feature $v \in R^m$, separate it into K mutually orthogonal subspaces.
- Why?: Forming an independent subspace structure that mirrors the common assumption of (approximately) independent FoV.

Contributions

- Proposes a continuous layer-centric disentanglement method.
- Provides a plug-and-play OSP layer for convolutional, linear, and Transformer-based architectures.
- Demonstrates improvements in disentanglement and downstream generalization across CV, NLP, and fine-tuning tasks.

Method



1) Orthogonal Subspace Projection

- Let $A' = QR$, $Q \in \mathbb{R}^{m \times m}$, $Q = [Q_1, Q_2, \dots, Q_k]$, $Q_i \in \mathbb{R}^{m \times d}$.
- Then $P_i = Q_i Q_i^T \in \mathbb{R}^{m \times m}$ is the orthogonal projection onto the d -dimensional subspace $S_i = \text{col}(Q_i)$.
- Any vector $v \in \mathbb{R}^m$ projected onto subspace S_i and S_j is orthogonal $P_i v \perp P_j v$.

2) Dimension Compression

- Generating the set of k projected vectors $\{p_i v\}_{i=1}^k$ **increase** memory usage.
- Projection matrix rank (d) is **lower** than output dimension (m), where $\text{rank}(P_i) = d \ll m$.
- Output feature $z = [z_1, z_2, \dots, z_k] \in \mathbb{R}^{kd} = \mathbb{R}^m$, where $z_i = Q_i^T v \in \mathbb{R}^d$.

3) Computational Efficiency

- $\hat{Q} = [Q_1 C, Q_2 C, \dots, Q_k C] = Q_{total}(I_k \otimes C)$.
- $C = (I_d - A)(I_d + A)^{-1}$, $A^T = A^{-1}$.

4) Plug-and-Play

- Linear and Convolutional Layers.

Results

Disentanglement Learning (DL)

- OSP improves disentanglement without explicit disentanglement objectives.
- OSP-VAE further improves disentanglement performance.
- Factor directions become closer to the ideal identity-matrix structure.

Dataset	Metric	AE	OSP-AE	Dataset	Model	β -VAE	FVM	MIG
3D Shapes	β -VAE	67.20(± 2.28)	76.80 (± 5.02)	3D Shapes	β -VAE	85.80(± 8.66)	75.86(± 10.73)	35.41(± 17.06)
	FVM	50.62(± 4.15)	60.73 (± 7.00)		OSP- β -VAE	90.60 (± 4.72)	78.28 (± 10.06)	43.33 (± 13.95)
	MIG	5.46(± 1.18)	8.87 (± 1.58)		β -TCVAE	88.80(± 5.02)	74.70(± 3.39)	34.27(± 12.54)
	SAP	2.23(± 0.23)	2.93 (± 0.49)		OSP- β -TCVAE	90.80 (± 5.22)	84.88 (± 4.64)	41.32 (± 14.06)
	DCI-Disc.	10.52(± 2.09)	14.66 (± 2.69)	MPI3D-toy	CLG-VAE	79.60(± 18.83)	68.80(± 18.13)	34.41(± 14.13)
MPI3D-toy	DCI-Com.	11.34(± 1.50)	16.26 (± 1.98)		OSP-CLG-VAE	85.80 (± 8.30)	78.68 (± 10.59)	34.79 (± 12.42)
	β -VAE	55.20(± 5.22)	61.33 (± 4.62)		β -VAE	37.20(± 16.39)	33.34(± 10.90)	6.91 (± 9.78)
	FVM	33.48(± 2.85)	37.67 (± 4.57)		OSP- β -VAE	43.80 (± 12.73)	38.20 (± 8.69)	5.83 (± 6.35)
	MIG	1.99(± 0.38)	2.65 (± 1.61)	MPI3D	β -TCVAE	34.40(± 15.19)	32.00(± 12.35)	6.42(± 11.09)
MPI3D	SAP	1.23(± 0.38)	1.40 (± 0.33)		OSP- β -TCVAE	46.00 (± 6.16)	37.53 (± 4.37)	17.36 (± 7.73)
	DCI-Disc.	7.70(± 1.00)	8.59 (± 2.27)		CLG-VAE	41.00(± 5.03)	34.97(± 1.04)	19.30(± 1.78)
	DCI-Com.	7.53(± 1.03)	8.18 (± 2.11)		OSP-CLG-VAE	48.00 (± 3.65)	39.86 (± 1.39)	19.90 (± 1.09)

- (1) AE vs. OSP-AE. (2) DL performance on 3D Shapes and MPI3D. (3) Orthogonality between factor directions.

Computer Vision (CV)

- (4-5) OSP improves CV classification in both scratch training and fine-tuning.
- (6) OSP-ResNet18 attends more to class-relevant object.

Models	Params	CIFAR-100	ImageNet	Models	Fine-Tuning	# of Param. ↓	ImageNet ↑
ResNet-18	11.69M	72.97 ± 1.38	64.44 ± 0.37	ResNet 18	Full fine-tuning	11.69M	66.69(± 0.06)
OSP-ResNet-18	11.70M	74.00 ± 0.35	65.71 ± 0.23		OPT (Liu et al., 2020)	46.50M	OOM
ResNet-50	25.56M	73.28 ± 0.61	69.86 ± 0.12		S-OPT	3.39M	67.74
OSP-ResNet-50	25.58M	74.64 ± 0.68	70.79 ± 0.14		OSP-fine-tuning	0.50M	50.07(± 3.08)
ViT-Base/16	87.40M	-	78.22 ± 0.45	ResNet 50	Full fine-tuning	25.56M	76.59(± 0.08)
OSP-ViT-Base/16	87.58M	-	80.33 ± 0.13		OSP-fine-tuning	1.00M	77.67 (± 0.08)

- (4) Training from scratch (5) Fine-tuning (6) Visualization

Natural Language Processing

OSP improves word analogy and text classification performance.

	SAT			U2			U4			Google			BATS			Avg.			
	ppl	mppl	pmi	ppl	mppl	pmi	ppl	mppl	pmi	ppl	mppl	pmi	ppl	mppl	pmi	ppl	mppl	pmi	
BERT	29.23	28.76	28.55	35.71	34.52	37.50	37.50	35.03	38.13	34.31	33.27	33.45	34.73	35.04	36.46	34.30	33.32	34.82	
OSP-BERT	30.43	28.33	29.97	39.29	39.29	36.90	34.58	36.67	36.25	34.52	34.89	34.52	34.76	34.44	34.43	34.71	34.73	34.42	
RoBERTa	30.51	29.03	29.86	36.76	34.33	35.71	34.43	35.21	35.83	33.06	33.41	32.83	33.25	33.25	33.87	33.60	33.04	33.86	
OSP-RoBERTa	29.95	30.19	31.99	39.48	37.90	37.90	38.75	38.33	38.96	33.21	32.85	33.33	33.77	33.97	34.16	33.66	35.07	34.69	35.26
		Param.	WNLI	RTE	STS-B	MRPC	CoLA	SST-2	QNLI	QQP	MNLI								
BERT	109.48M	39.44	60.29	86.29/86.08	81.37/87.21	53.91	92.32	90.66	90.51/87.31	83.91/84.10									
OSP-BERT	109.63M	46.48	65.89	86.38/86.40	81.62/87.44	53.39	92.43	91.16	90.60/87.43	84.52/84.48									
BERT-Large	335.14M	56.33	63.54	87.38/87.49	81.62/87.09	57.82	92.78	92.26	91.31/88.40	86.34/86.44									
OSP-BERT-Large	335.40M	56.33	94.62	88.36/88.38	82.60/87.80	58.59	93.00	92.22	91.42/88.50	86.68/86.75									
RoBERTa-large	355.43M	56.34	72.20	91.09/91.17	87.99/91.51	64.07	95.99	94.33	91.75/89.09	90.74/90.48									
OSP-RoBERTa-large	355.76M	56.34	72.20	91.25/91.34	88.73/91.90	63.99	96.33	94.42	91.91/89.30	90.98/90.76									